

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use your Driving Question Board, your cumulative model poster and your notes from Lessons 1 to 7. Write your answers in the space under each prompt. Show your working clearly, use the units m, m², KES and s, and check every solution in its context. Round decimals to 2 d.p. in final answers. Write in full sentences where a prompt asks you to explain.

Part 1: Why the committee got stuck

The Githurai committee drew a sketch on manila paper: a rectangular hall with width = w , length = $w + 2$ and Area = 120 m². (a) Write the area as an equation in w and expand it into the form $ax^2 + bx + c = 0$. State a , b and c . (b) The committee also tried the perimeter $2(w + w + 2)$. Write it in simplest form. Compare it with the area equation: which one is linear and which one is quadratic, and why does this explain why the committee could not find w using a linear equation? (c) Look at the trial-and-error table below and explain why guessing is not a reliable method.

w (m)	Length (m)	Area (m ²)
8	10	80

(a) Area = $w(w + 2) = 120$, so $w^2 + 2w = 120$, which gives $w^2 + 2w - 120 = 0$. In the general form $ax^2 + bx + c = 0$, $a = 1$, $b = 2$ and $c = -120$. (b) Perimeter = $2(w + w + 2) = 4w + 4$. This is linear because the highest power of w is 1. The area equation is quadratic because the highest power of w is 2. The area has w multiplied by w , which gives a w^2 term, so a linear equation cannot solve it. The perimeter $4w + 4$ is linear but it does not give the area. (c) The table shows that $w = 10$ gives exactly 120 m², but we only found it because we tried that value. The areas 80, 99, 120, 143 and 168 grow by different amounts, so there is no simple pattern to predict the answer. Guessing works only if we are lucky, and it would not work if the answer were not a whole number. We need a method that works for quadratic equations.

9	11	99
10	12	120
11	13	143
12	14	168

Part 2: Expanding and factorising

(a) A Kericho tea farmer fences a plot of length $(x + 5)$ m and width $(x + 3)$ m. Expand the area $(x + 5)(x + 3)$ using the distributive law or FOIL, and write the fencing perimeter in simplest form. Which of the two has an x^2 term, and why? **(b)** Expand $(2x - 1)(3x + 4)$. **(c)** Factorise the hall expression $w^2 + 2w - 120$ into two brackets. Explain how you found the two numbers you needed, and show how you can check your answer. **(d)** Explain how expanding and factorising are connected.

(a) Area = $(x + 5)(x + 3) = x^2 + 3x + 5x + 15 = x^2 + 8x + 15$ m². Perimeter = $2[(x + 5) + (x + 3)] = 4x + 16$ m. The area has an x^2 term because one length that contains x is multiplied by another length that contains x . The perimeter only adds lengths, so it stays linear. **(b)** $(2x - 1)(3x + 4) = 6x^2 + 8x - 3x - 4 = 6x^2 + 5x - 4$. **(c)** We need two numbers with product -120 and sum 2. The pair 12 and -10 works because $12 \times (-10) = -120$ and $12 + (-10) = 2$. So $w^2 + 2w - 120 = (w + 12)(w - 10)$. Check by expanding: $w^2 - 10w + 12w - 120 = w^2 + 2w - 120$, which matches. **(d)** Factorising is the reverse of expanding. Expanding turns a product of brackets into $ax^2 + bx + c$, and factorising writes $ax^2 + bx + c$ as a product of brackets again. We can check any factorisation by expanding it.

Part 3: Solving the hall equation three ways

(a) Use the Zero Product Property to solve $(w + 12)(w - 10) = 0$. State the property in your own words. **(b)** Which solution can be the width of the hall, and why? State the width and length of the hall and check the area. **(c)** Solve $w^2 + 2w - 120 = 0$ using the quadratic formula. Show the discriminant $D = b^2 - 4ac$. **(d)** Complete the

(a) Zero Product Property: if $A \times B = 0$ then $A = 0$ or $B = 0$. So $w + 12 = 0$ or $w - 10 = 0$, giving $w = -12$ or $w = 10$. **(b)** A width cannot be negative, so we reject $w = -12$. The hall has width 10 m and length 12 m. Check: $10 \times 12 = 120$ m², which matches the target. This also matches the $w = 10$ row of the trial-and-error table. **(c)** $a = 1$, $b = 2$, $c = -120$. $D = 2^2 - 4(1)(-120) = 4 + 480 = 484$, and the square root of 484 is 22. $w = (-2 \pm 22)/2$, so $w = 20/2 = 10$ or $w = -24/2 = -12$. $D > 0$, so there are two real solutions. **(d)** $w^2 + 2w = 120$. Add $(2/2)^2 = 1$ to both sides: $w^2 + 2w + 1 = 121$, so $(w + 1)^2 = 121$. Taking square roots gives $w + 1 = 11$ or $w + 1 = -11$, so $w = 10$ or $w = -12$. **(e)** The three methods (factorisation, completing the square and the quadratic formula) all give $w = 10$ or $w = -12$. Context then tells us to use $w = 10$, so the hall is width 10 m, length 12 m.

square on $w^2 + 2w = 120$ to show that the answer is the same. (e) What does the fact that all three methods agree tell you?

Part 4: Choosing a method and using context

The Nakuru school garden club has a plot with width x m, length $(x + 6)$ m and area 48 m², so $x^2 + 6x = 48$. (a) Explain why factorisation is not a good method here. (b) Solve it by completing the square and give the width and length to 2 d.p. (c) Mama Wanjiru sells mandazi and wants revenue of KES 700 a day. Her equation is $p^2 - 30p + 175 = 0$, where p is the price in KES. Use the discriminant to find the prices p to 2 d.p. Both prices give the revenue she wants. Which price should she choose, and why? (d) In both problems, how did context help you decide which solution to use?

(a) We need two integers with product -48 and sum 6, and no such pair exists, so the equation does not factorise. (b) $x^2 + 6x = 48$. Add $(6/2)^2 = 9$ to both sides: $x^2 + 6x + 9 = 57$, so $(x + 3)^2 = 57$. Then $x + 3 = \pm \sqrt{57}$, so $x = -3 \pm \sqrt{57}$. $\sqrt{57} = 7.55$, so $x = 4.55$ or $x = -10.55$. A width cannot be negative, so the width is 4.55 m and the length is $4.55 + 6 = 10.55$ m. Check: $4.55 \times 10.55 = 48$ m² (approximately). (c) $a = 1$, $b = -30$, $c = 175$. $D = 900 - 700 = 200$, $\sqrt{200} = 14.14$. $p = (30 \pm 14.14)/2$, so $p = 7.93$ or $p = 22.07$ KES. $D > 0$, so there are two real solutions. Both give revenue of KES 700, but at 7.93 KES she sells about 88 mandazi and at 22.07 KES only about 32. She should choose 7.93 KES because she sells more mandazi and serves more customers for the same revenue. (d) In the garden, context made us reject the negative width -10.55. In the mandazi problem, both solutions were possible, so context (selling more mandazi) helped us choose the better price. A quadratic equation can have two solutions, and the context decides which one to use.

Part 5: Final explanation

Answer the driving question in a short written explanation (8 to 10 sentences): How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Your explanation must (1) say how quadratic equations differ from linear ones and why they are necessary, (2) describe how the

Quadratic equations have an unknown multiplied by itself, so the highest power is 2, while a linear equation such as the perimeter $4w + 4$ has highest power 1. We need quadratic equations when a quantity like area is a product of two lengths that both contain the unknown, as in the Githurai hall. The committee knew that the width is w , the length is $w + 2$ and the area is 120 m², so $w(w + 2) = 120$, which becomes $w^2 + 2w - 120 = 0$. Factorisation, $(w + 12)(w - 10) = 0$, and the Zero Product Property gave $w = 10$ or $w = -12$, and the quadratic formula with $D = 484$ gave the same solutions. The discriminant tells us the number of real solutions: $D > 0$ gives two, $D = 0$ gives one repeated solution and $D < 0$ gives none. A width cannot be negative, so context makes us reject $w = -12$, and the hall has width 10 m and length 12 m. Quadratic thinking also helps with other planning problems. For example, Mama Wanjiru's mandazi equation $p^2 - 30p + 175 = 0$ has $D = 200$, so it gives two prices, 7.93 KES and 22.07 KES. Both give revenue of KES 700, but context tells her to choose 7.93 KES because she sells more mandazi. So quadratic equations let our community find unknown dimensions and prices for real plans.

hall problem was turned into an equation and solved, (3) explain what the discriminant D tells us ($D > 0$, $D = 0$ and $D < 0$) and how context is used to choose a solution, and (4) give one other community problem from Part 4 where quadratic thinking helps, and say how context decided the answer.	
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Distinguishing quadratic from linear and forming the equation	Correctly writes $w(w + 2) = 120$ as $w^2 + 2w - 120 = 0$, identifies $a = 1$, $b = 2$ and $c = -120$, and clearly explains using $w \times w$ and the linear perimeter $4w + 4$ why a linear method fails.	Forms the equation and identifies a , b and c correctly, with a partial explanation of the difference between quadratic and linear.	Has errors in the equation or in a , b and c , or gives no explanation of why the equation is not linear.
Expanding and factorising accurately	Expands $(x + 5)(x + 3)$ and $(2x - 1)(3x + 4)$ correctly, factorises $w^2 + 2w - 120$ as $(w + 12)(w - 10)$, and checks by expanding.	Expands and factorises mostly correctly, with a minor sign or collecting-terms error or no check.	Several errors in expanding or factorising, and no connection made between the two.
Solving by the three methods	Solves correctly by the Zero Product Property, completing the square and the quadratic formula, including $D = 484$, and shows the methods agree. Solves the Nakuru and mandazi problems correctly to 2 d.p.	Solves correctly by at least two methods, with minor slips in the working or rounding.	Can solve by one method only, or makes major errors in the formula, the discriminant or the square roots.
Using context and the discriminant to judge solutions	Rejects $w = -12$ with a clear reason, states width 10 m and length 12 m with a check of 120 m ² , explains $D > 0$, $D = 0$ and $D < 0$ correctly, and justifies the mandazi price choice.	Rejects the impossible solution and states the answer, with a limited explanation of D or of the price choice.	Gives both solutions without choosing, or gives no reason for rejecting one.
Final explanation and communication	Gives a clear, connected explanation answering the driving question with the hall data, correct terms (solutions, discriminant, Zero Product Property) and another community example.	Explanation covers most required points with mostly correct terms but limited links.	Explanation is incomplete or does not use evidence from the lessons.

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